

# CSE490 (D)

## INTRODUCTION TO FOOTBALL DATA AND ANALYTICS

IN ASSOCIATION WITH BANGLADESH FOOTBALL FEDERATION

SPRING 2026



- Professional Certificate from BFF
- Internship Opportunity at **BFF Data Science** Department after successful completion
- Learning from the **industry experts** of Football Data Science
- Opportunity to **co-author** research papers for **Sports Data Conferences**

# **CSE490 (D)**

# **Introduction to**

# **Football Data & Analytics**

## **Lecture-1**

Prepared by-  
*Tarvir Anjum Aditto*  
Lecturer,  
CSE, Brac University



# Course Objectives

- Introduce research-oriented football analytics using computing and statistical methods
- Develop analytical, modeling, and visualization skills using real-world data
- Foster independent research, critical thinking, and lifelong learning
- Strengthen ethical reasoning, teamwork, and professional communication



# Course Content

| Module                                | Lessons                                                                                    |
|---------------------------------------|--------------------------------------------------------------------------------------------|
| Data Foundations & Visualization      | Introduction to Event Data & KPIs; Visualizing Football; Passing Networks & Team Structure |
| Statistical Models & Expected Goals   | Choosing & Fitting Models; Advanced Models; Building an xG Model                           |
| Player & Action Valuation             | Scouting and Player Profiles; Valuing Specific Actions                                     |
| Uncertainty, Prediction & Trends      | Randomness in Football; Team-Level KPIs & Trend Analysis                                   |
| Space Control & Player Movement       | Pitch Control Tools; Off-Ball Movement & Positioning                                       |
| Physical Data & Clustering Techniques | Physical Performance Metrics; Clustering in Football Analytics                             |

# Assessment Structure

| Assessment            | Weight |
|-----------------------|--------|
| Quiz                  | 10%    |
| Midterm               | 20%    |
| Research Project      | 30%    |
| Mini Case Study       | 10%    |
| Final                 | 20%    |
| Research Presentation | 10%    |

# Research Project

## Nature

- Individual or group (max 3 students)
- Dataset: [StatsBomb](#) / [Wyscout](#) / Open football data

## Deliverables

- ❖ Research proposal
- ❖ Literature review
- ❖ Methodology & modeling
- ❖ Results & discussion
- ❖ Ethical considerations
- ❖ Final report & presentation



# Research Project Structure

| Assessment            | Weight |
|-----------------------|--------|
| Problem Formulation   | 15%    |
| Literature Review     | 15%    |
| Methodology           | 20%    |
| Analysis & Validation | 25%    |
| Ethics & Limitation   | 10%    |
| Communication         | 15%    |



# Tools

- Python
- Open datasets: [StatsBomb](#) / [WyScout](#)
- Libraries: pandas, numpy, matplotlib, [mplsoccer](#), statsbombpy
- ***N:B:*** We will mostly use google colab, but it's better if you have conda/ VScode installed





# Statsbomb (Data structure)

**Follow this notebook-**

<https://colab.research.google.com/drive/11izlsskxXn91-AVLfADYO0GOR1sFivDd?usp=sharing>

**Data Structure:** *Competition* → *Season* → *Match* → *Events*

**How to access data:**

- ☐ StatsBomb assigns unique IDs at each level
- ☐ Each competition, season, and match has its own ID
- ☐ Example
  - Competition: FIFA World Cup, competition\_id = 43
  - Season: 2018, season\_id = 106
  - Match: England vs Colombia, match\_id = 7580

# Statsbomb (Competitions)

## Loading competitions as a DataFrame:

- Provides high-level information such as:
  - Competition name / country
  - Season
  - Number of matches
  - Gender
  - Availability of **360 data**

| #  | Column                    | Non-Null | Count    | Dtype  |
|----|---------------------------|----------|----------|--------|
| 0  | competition_id            | 75       | non-null | int64  |
| 1  | season_id                 | 75       | non-null | int64  |
| 2  | country_name              | 75       | non-null | object |
| 3  | competition_name          | 75       | non-null | object |
| 4  | competition_gender        | 75       | non-null | object |
| 5  | competition_youth         | 75       | non-null | bool   |
| 6  | competition_international | 75       | non-null | bool   |
| 7  | season_name               | 75       | non-null | object |
| 8  | match_updated             | 75       | non-null | object |
| 9  | match_updated_360         | 57       | non-null | object |
| 10 | match_available_360       | 11       | non-null | object |
| 11 | match_available           | 75       | non-null | object |

# Statsbomb (Matches)

## Loading matches for a competition as a DataFrame:

- Provides match-level information such as:
  - Home team & away team
  - Match score
  - Stadium
  - Match stage
  - Referee
  - Team managers
  - Match date

| #  | Column                | Non-Null Count | Dtype  |
|----|-----------------------|----------------|--------|
| 0  | match_id              | 64 non-null    | int64  |
| 1  | match_date            | 64 non-null    | object |
| 2  | kick_off              | 64 non-null    | object |
| 3  | competition           | 64 non-null    | object |
| 4  | season                | 64 non-null    | object |
| 5  | home_team             | 64 non-null    | object |
| 6  | away_team             | 64 non-null    | object |
| 7  | home_score            | 64 non-null    | int64  |
| 8  | away_score            | 64 non-null    | int64  |
| 9  | match_status          | 64 non-null    | object |
| 10 | match_status_360      | 64 non-null    | object |
| 11 | last_updated          | 64 non-null    | object |
| 12 | last_updated_360      | 64 non-null    | object |
| 13 | match_week            | 64 non-null    | int64  |
| 14 | competition_stage     | 64 non-null    | object |
| 15 | stadium               | 64 non-null    | object |
| 16 | referee               | 64 non-null    | object |
| 17 | home_managers         | 64 non-null    | object |
| 18 | away_managers         | 64 non-null    | object |
| 19 | data_version          | 64 non-null    | object |
| 20 | shot_fidelity_version | 0 non-null     | object |
| 21 | xy_fidelity_version   | 0 non-null     | object |

# Statsbomb (Events)

## Loading events for a match as a DataFrame:

- Each row represents **one on-ball action**
- Examples of events:
  - Pass
  - Shot
  - Foul
  - Duel
  - Clearance

## Information stored in event data includes:

- Event type
- Time (minute, second)
- Event location on the pitch
- Player and team involved
- Event-specific details (e.g., pass type, shot outcome)



# Statsbomb (Events)

## Important observation:

- Most columns contain **NaN values**
- This is expected because:
  - Each event type only uses a **subset of columns**
  - Columns not relevant to an event remain empty

Data columns (total 83 columns):

| #  | Column                         | Non-Null Count | Dtype   |
|----|--------------------------------|----------------|---------|
| 0  | 50_50                          | 4 non-null     | object  |
| 1  | bad_behaviour_card             | 3 non-null     | object  |
| 2  | ball_receipt_outcome           | 160 non-null   | object  |
| 3  | ball_recovery_recovery_failure | 13 non-null    | object  |
| 4  | block_deflection               | 2 non-null     | object  |
| 5  | block_offensive                | 1 non-null     | object  |
| 6  | carry_end_location             | 851 non-null   | object  |
| 7  | clearance_aerial_won           | 4 non-null     | object  |
| 8  | counterpress                   | 108 non-null   | object  |
| 9  | dribble_nutmeg                 | 2 non-null     | object  |
| 10 | dribble_outcome                | 29 non-null    | object  |
| 11 | duel_outcome                   | 27 non-null    | object  |
| 12 | duel_type                      | 62 non-null    | object  |
| 13 | duration                       | 2411 non-null  | float64 |
| 14 | foul_committed_advantage       | 2 non-null     | object  |
| 15 | foul_committed_card            | 5 non-null     | object  |
| 16 | foul_committed_penalty         | 1 non-null     | object  |

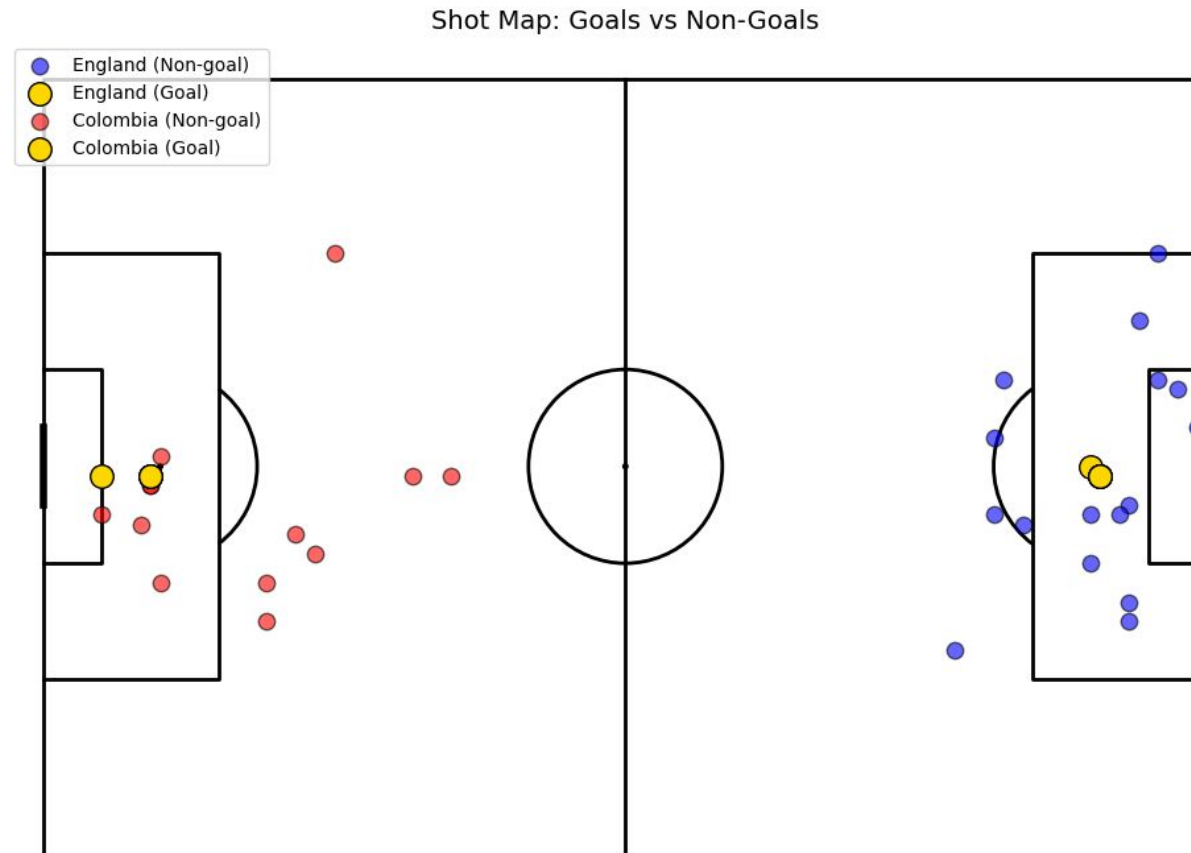
# Statsbomb (Events= Shots)

We can extract very specific info (Non-NaN) of a “shot” event

| Event Type | Player          | Team     | Minute | Location (x, y) | Body Part  | Shot Type | Outcome    | xG     |
|------------|-----------------|----------|--------|-----------------|------------|-----------|------------|--------|
| Shot       | Ashley Young    | England  | 5      | (115.0, 18.0)   | Right Foot | Free Kick | Saved      | 0.0098 |
| Shot       | Raheem Sterling | England  | 7      | (112.0, 54.0)   | Right Foot | Open Play | Blocked    | 0.0382 |
| Shot       | Raheem Sterling | England  | 12     | (98.0, 37.0)    | Right Foot | Open Play | Blocked    | 0.0451 |
| Shot       | Harry Kane      | England  | 15     | (119.0, 36.0)   | Head       | Open Play | Off Target | 0.6251 |
| Shot       | Juan Cuadrado   | Colombia | 21     | (97.0, 56.0)    | Right Foot | Open Play | Blocked    | 0.0218 |

# Statsbomb (Visualizing shots)

We can draw standard size football pitch and visualize events using **mplsoccer's Pitch**





# Statsbomb (360 Data)

Statsbomb offers 360 data which track not only location of an event but also players' location.

| # | Column       | Non-Null Count | Dtype  |
|---|--------------|----------------|--------|
| 0 | id           | 41562 non-null | object |
| 1 | visible_area | 41562 non-null | object |
| 2 | match_id     | 41562 non-null | int64  |
| 3 | teammate     | 41562 non-null | bool   |
| 4 | actor        | 41562 non-null | bool   |
| 5 | keeper       | 41562 non-null | bool   |
| 6 | location     | 41562 non-null | object |

*360 frame info*

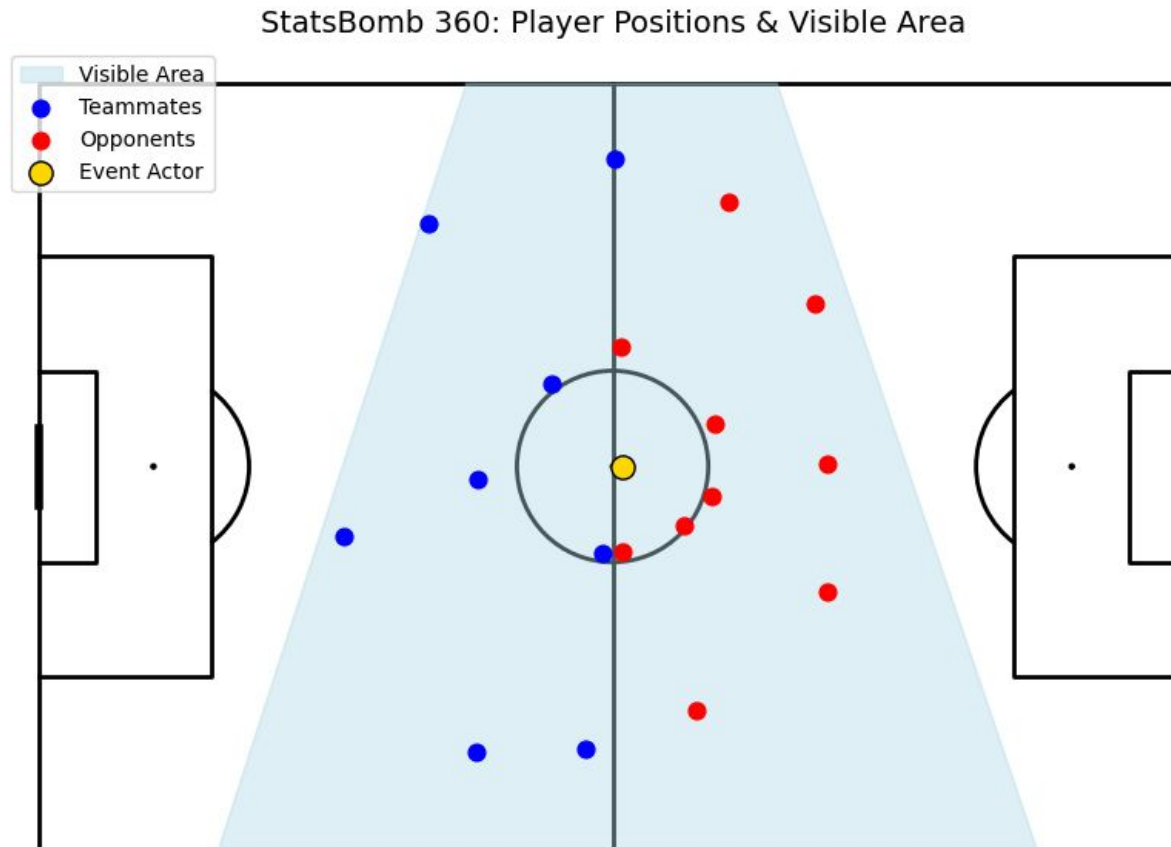
| Player locations at the moment of the event: |           |           |          |       |        |
|----------------------------------------------|-----------|-----------|----------|-------|--------|
|                                              | x         | y         | teammate | actor | keeper |
| 0                                            | 31.885449 | 47.289130 | True     | False | False  |
| 1                                            | 40.762566 | 14.657225 | True     | False | False  |
| 2                                            | 45.765232 | 69.890950 | True     | False | False  |
| 3                                            | 45.923999 | 41.324219 | True     | False | False  |
| 4                                            | 53.649592 | 31.368630 | True     | False | False  |
| 5                                            | 57.221077 | 69.484720 | True     | False | False  |
| 6                                            | 58.963536 | 49.056717 | True     | False | False  |
| 7                                            | 60.159949 | 7.849366  | True     | False | False  |
| 8                                            | 60.952056 | 27.553933 | False    | False | False  |

*Player locations at the moment of a particular event*



# Statsbomb (360 Data)

What can you observe from this pitch image?



Serbia vs Switzerland

Match date: 2022-12-02

Serbia starting formation: 3412

Switzerland starting formation: 4231

***These are also extracted by using match\_id  
and keys like home\_team, away\_team,  
Match\_date, type, teactics***

# Event Data vs Tracking Data

## Event Data

- Records **on-ball actions** (passes, shots, tackles, fouls)
- Logged **only when an event occurs**
- Sparse data, many **NaN columns** (not all fields apply to all events)
- Used for **xG, shot maps, pass maps, KPIs**

## Tracking Data

- Records **continuous positions** of all players & the ball
- Collected **every frame** (e.g, 10 times/sec)
- Dense, large datasets
- Used for **pressing, team shape, space control, movement analysis**



# References

- **Soccermatics Blog:** [Welcome to Soccermatics — Soccermatics documentation](#)
- [Course: mathematical modelling of football - YouTube](#)
- **Book: Soccermatics by David Sumpter**

